

ShopSense AI

From Surveillance to Intelligence



TEAM INNOV8

Team Size: 4 | AI-Powered Retail Intelligence Platform

Cameras Watch Everything. They Understand Nothing.

Retail stores struggle to manage customer movement, staff availability and in-store operations — leaving customers waiting and managers flying blind.



Wasted Time

Customers spend unnecessary time searching for products or assistance.



Staff Misallocation

Available staff aren't where they're needed — service gets delayed.



Zero Visibility

Managers lack real-time insight into movement and staff availability.



Underused CCTV

Existing cameras serve security only — not customer experience.



No Universal Fix

Every store has a different layout, so fixed solutions fall short.



Manual & Unscalable

Monitoring flow and staffing by hand can't scale across stores.

CORE QUESTION *How can existing store cameras become an intelligent system that connects customers with available staff — without new infrastructure?*

One Idea: Turn Existing Cameras Into an Intelligent Layer

ShopSense AI is an AI-powered retail assistance and store-optimization system.

Instead of using CCTV only for surveillance, it applies computer vision and AI analytics to understand what is happening inside the store in real time.

It detects when a customer may need assistance and, if an appropriate staff member is available, intelligently notifies them — closing the loop between need and response.



The system reads customer movement, store activity, and staff position — then acts on it.

No new hardware. Just new intelligence.

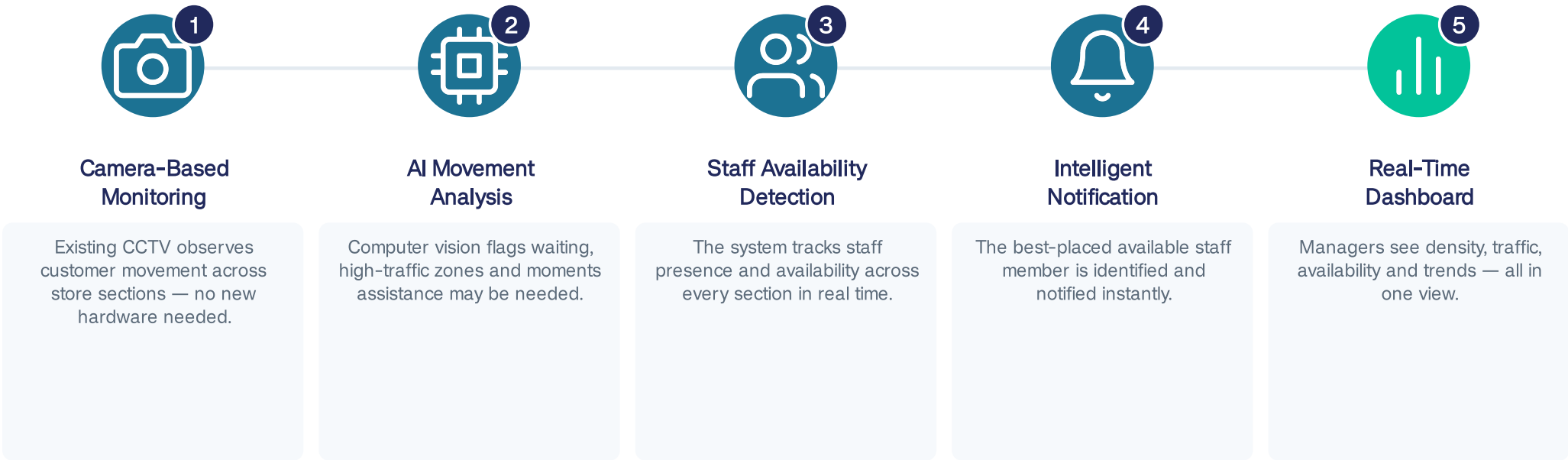


Layout-Adaptive by Design

Configures to each store's sections, cameras and staff zones.

How ShopSense AI Works

A five-stage loop — from watching the store to guiding the right person to the right place.



The loop repeats continuously — turning every camera into a live operational sensor.

What Makes ShopSense AI Different



Traditional CCTV

“Something is happening.”

Passive footage for security review only. No understanding of who, where, or what to do next.



ShopSense AI

“What, where — and who can respond.”

Passive footage becomes an active decision-support system for retail operations.



Layout-Adaptive, Not One-Size-Fits-All

Every store has a different floor plan, product arrangement and staff count. During deployment, a store configures its own sections, cameras, staff zones and operational parameters — so ShopSense AI adapts to the environment instead of assuming every store is the same.

Proposed Technology Stack

Built on the store's existing camera infrastructure — layered with computer vision, real-time analytics and smart notifications.



Computer Vision

Reads customer movement and store activity from existing camera feeds.



AI / ML Analytics

Detects wait patterns, traffic hotspots and assistance -need signals.



Edge - Cloud Processing

Hybrid pipeline balances on-site speed with cloud-scale analysis.



Staff-Matching Engine

Matches an assistance need to the nearest available employee.



Notification Layer

Delivers real-time alerts to staff through the ShopSense interface.



Manager Dashboard

Web-based visualization of density, availability and trends.

Who Benefits, and How



Customers

- Faster, easier help when it's needed
- Less time spent searching, more time shopping



Store Staff

- Directed to where they're actually needed
- Less idle time, better-used effort



Store Managers

- Real-time visibility into flow & availability
- Data-driven staffing and layout decisions

Overall, ShopSense AI Aims To:



Reduce Wait Times



Improve Staff Utilization



Elevate Customer Experience



Surface Real-Time Insights

Where ShopSense AI Goes Next

From surveillance to intelligence.

From waiting to assistance.

From raw camera feeds to actionable retail insights.



Predictive Analytics

Anticipate high-traffic moments and staffing needs before they happen.



Multi-Store Scaling

Extend the layout-adaptive model across chains and formats.



Deeper Integration

Connect with inventory and POS systems for fuller operational context.



Personalized Assistance

Tailor in-store support to individual customer needs over time.